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QUIZ # 01

AI

05

Agent:-

An agent is anything that can perceive its environment and act on it.

Core Role: Bridge between environment and intelligent behavior. (visual diagram, - Environment - Agent box - Environment)

Agent & Environment Relationship:-

The agent gets input (percepts) from its environment and takes actions that affect it.

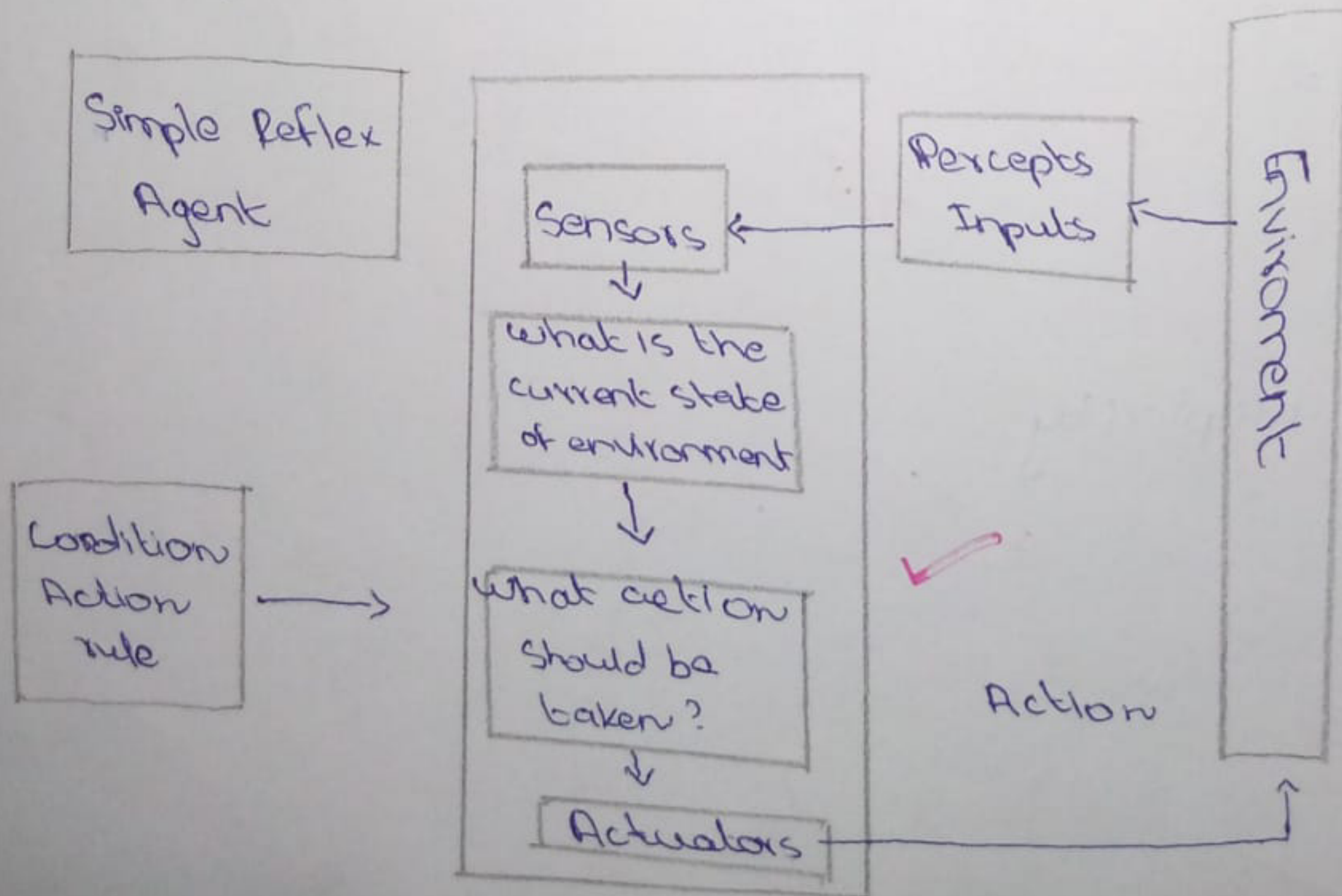
Cycle:

Perception → Decision → Agent → Environment

Example:-

A cleaning robot sees dirt (perception), moves to it (action)

Agent Diagram:-

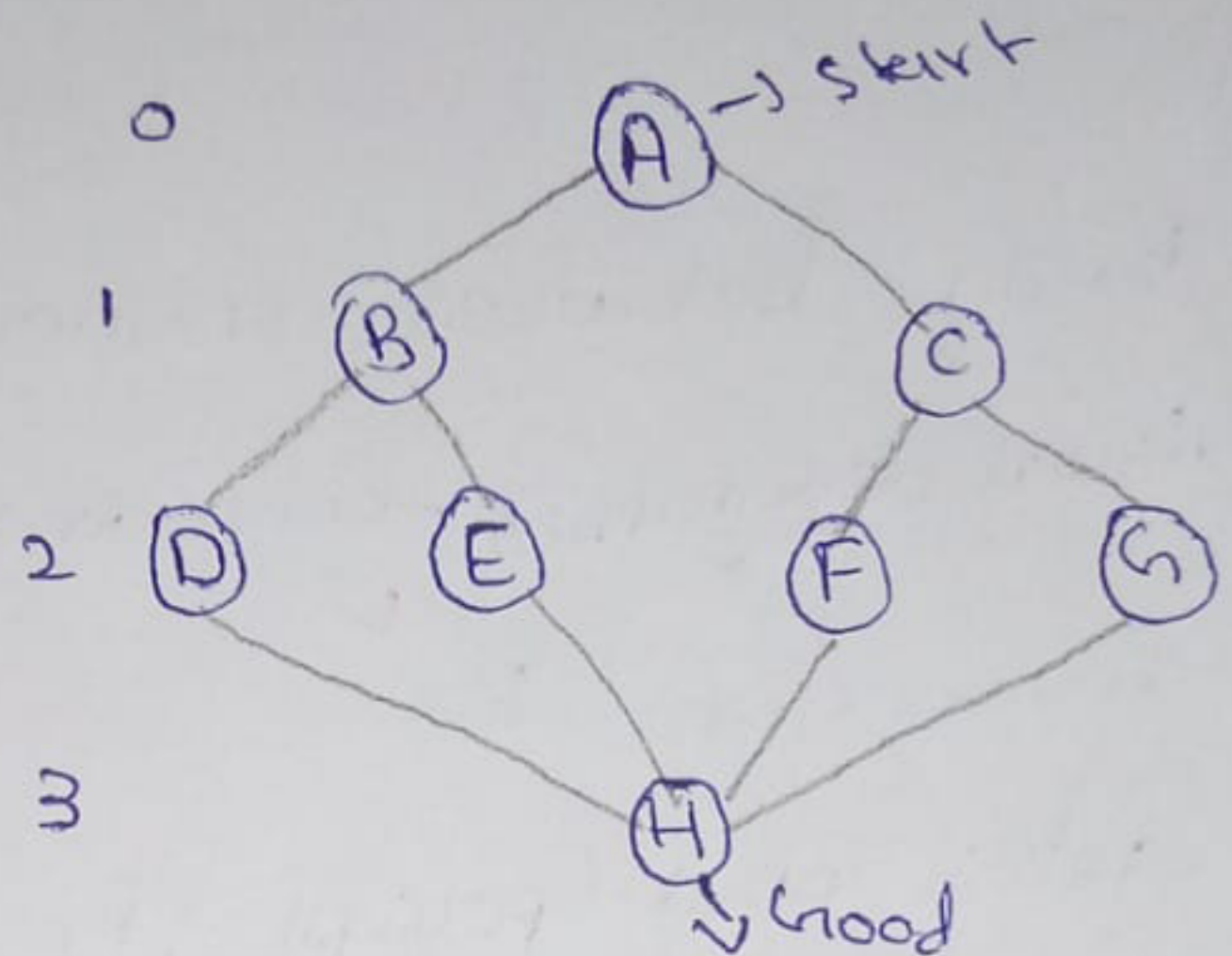


Breath First Search (BFS) :-

- It follows the principle of Queue (FIFO)
- Time & space complexity $O(b^d)$

STEPS:-

~~A~~
~~BC~~
~~CDE~~
~~DEFG~~
~~EFGH~~
~~FGH~~
~~GH~~
H



PATH: A → B → C → D → E → F → G → H

Path from node A to H: (all possible shortest path) :-

A → B → D → H

A → B → E → H

A → C → F → H

A → C → G → H

Complexity:-

$O(b^d)$

$O(2^3)$

$O(8)$ = Time Complexity

2.5